Qihan Shan

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EDUCATION

University of Pennsylvania Philadelphia, PA May 2025 – May 2027 (expected)
M.S. in Mechanical Engineering and Applied Mechanics — Mechatronics & Robotic Systems
Coursework: Robot Motion Control, Path Planning, Robot Kinematics, Machine / Deep / Reinforcement Learning

Zhejiang University–UIUC Institute (ZJUI) Sep 2021 – Jun 2025
B.S. in Mechanical Engineering, minor in Electrical Engineering | GPA 3.87 / 4.00

EXPERIENCE

AgiBot — Motion Control Algorithm Intern Shenzhen, China Jun 2026 – Present
X-Lab, Motion Intelligence Group | Layered navigation stack for the X2 humanoid

Semantic Planning — DualVLN reproduction with self-collected data (simulation)

- Built the VLN data pipeline end to end: collected 4,229 R2R/RxR trajectories across 60 Matterport3D scenes, projected them onto egocentric views with depth-based visibility filtering, and segmented them into 251,916 training samples (pixel-goal grounding, view adjustment, STOP).
- LoRA-finetuned System 2 (Qwen-VL-2.5-7B, where the paper fully finetunes) and trained the matching flow-matching diffusion-transformer System 1, conditioned on latent goal queries and high-frequency RGB.
- VLN-CE RxR val-unseen: SR 59.3 / SPL 51.6 — level with the 59.6 / 51.8 obtained by re-running the official weights in the same evaluation setup, at markedly lower data and GPU-memory cost.

Local Execution — Perceptive Forward-Dynamics Navigation (deployed on hardware)

- Migrated a learned perceptive Forward Dynamics Model (RSS 2025) to the X2 humanoid: proprioceptive history and an elevation map predict 5 s of SE(2) poses and failure risk, feeding an MPPI planner that emits velocity commands with no hand-tuned cost maps. Deployed the full pipeline on hardware — doorways, stairs, pillars, corridors — reaching 4 m goals within 0.3 m.
- Extended it to moving obstacles, a case the paper lists as untested: 80% avoidance of a 1 m/s obstacle under a 10 Hz replanning loop, the other 20% stopping to wait rather than colliding — no collisions.
- Substituted FDM for System 1 as the action expert, projecting System 2's pixel goals into body-frame waypoints — the path that carried language-driven navigation onto the real robot.

Perceptive Locomotion — Attention-Based Map Encoding, AME-2 reproduction (Isaac Lab)

- Reproduced attention-based map encoding, uncertainty-aware neural elevation mapping and teacher–student distillation (PPO, action distillation, representation alignment).
- Transferred from the paper's quadruped and point-foot biped to the armed X2 humanoid: 90% teacher / 75% student success on unseen terrain (95.2% / 82.4% on the paper's quadruped), a distillation gap in line with the original; fixed foothold drift on stair ascent caused by odometry–elevation-map misalignment.

PROJECTS

UAV 3-D Path Planning and Real-Flight Deployment Feb – May 2026
UPenn | Advisor: Prof. M. Ani Hsieh

- Designed an SE(3) rigid-body pose controller with a low-level PD attitude loop, and combined A* global planning with piecewise quintic-polynomial fitting for smooth, collision-free, dynamically feasible trajectories.
- Integrated the full planning-to-control stack on hardware and flew autonomously through a real 3-D maze to a commanded landing point.

Quadruped Loco-Manipulation with a Mounted Arm Sep 2024 – Jan 2025
ZJU-UIUC Physical Intelligence Lab | Advisor: Prof. Hua Chen

- Led whole-body control modeling for dynamic mobile grasping, handling body posture, foot contacts, end-effector pose and stability constraints within a single policy.
- Trained PPO policies in massively parallel simulation with rewards spanning locomotion stability, manipulation, and coordination between the two.

SKILLS

Programming: Python, C, C++, MATLAB
Learning & Simulation: PyTorch, Isaac Lab / Isaac Sim, Habitat, MuJoCo; PPO, LoRA/PEFT, teacher–student distillation, diffusion policy and flow matching
Algorithms & Deployment: MPPI sampling-based planning, learned forward dynamics models, elevation mapping, VLN data pipelines, sim-to-real onboard deployment
Tools: ROS2, Linux, Git, SolidWorks, KiCad
Awards & Languages: Second Prize, National RoboMaster Robotics Competition 3v3, 2023 (ZJUI Meta RoboMaster Team) | Mandarin (native), English (TOEFL 103)